

Grades

 Print

| Grade Item   | Points     | Weight Achieved | Grade | Comments and Assessments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|------------|-----------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assignment 1 | 17.75 / 25 | 7.1 / 10        | B     | <div>Overall Feedback</div> <div>total score: 17.75/29</div> <div>You clearly put in a lot of effort and show strong intuition and engagement with the material, which is great to see. To improve your score, try to focus on precision: state the correct model, notation, and final result explicitly and concisely before adding explanations, as this will help your good understanding translate into full credit.</div> <div>-----</div> <div>Deduction details:</div> <div><b>Q1(a) -0.5:</b> You eventually reach the correct SSE/OLS objective, but the likelihood/log-likelihood derivation is confused (you mix up <math>p(x)</math> vs <math>p(y x,w)</math>, mention integrating over <math>x,w</math>, and the steps are not a clean “log-likelihood → drop constants → SSE”).</div> |

**Q1(b) -0.5:** You mention i.i.d. Gaussian noise, but you don't clearly state the key assumption that **X is fixed (or independent of the noise)**, and you don't clearly state Bayesian LR assumptions beyond "posterior  $\propto$  prior  $\times$  likelihood" (missing an explicit prior statement / clarity).

**Q2(a) -0.25:** Correct main conclusion that  $\Sigma_N$  shrinks as  $X^T X$  grows, but the explanation is a bit muddled (e.g., "non-proportional to N" phrasing) and doesn't clearly mention shrinkage in informative eigen-directions.

**Q3(d) -0.5:** You give generic covariance definitions, but you don't explain the key point: it is the **projection of parameter uncertainty through  $\mathcal{X}^*$** , so it depends on the direction/coverage of  $\mathcal{X}^*$ .

**Q3(e) -0.5:** You describe MLE vs MAP generally, but you do **not** clearly make the required contrast: plug-in prediction treats  $\mathcal{W}_w$  as fixed and **ignores the epistemic term** (so variance doesn't inflate in poorly learned regions).

**Q4(a) -0.25:** Mostly right (narrow in data-rich, wide in data-poor), but you add some misleading "overfitting  $\rightarrow$  variance to 0" language; the key point is widening away from observed regions.

**Q4(b) -0.25:** Partially right, but you don't really connect to the **geometry**:  $\mathcal{X}^T \Sigma_N \mathcal{X}$  is small near well-spanned/covered directions and larger in poorly sampled directions.

**Q5(a) -0.5:** You propose "entropy-based uncertainty sampling," but the expected simple answer is: choose  $\mathcal{X}^*$  that **maximizes  $\beta^{-1} + \mathcal{X}^{*T} \Sigma_N \mathcal{X}^*$**  (or just the epistemic term).

**Q5(b) -0.5:** The justification should be "this targets directions that shrink  $\Sigma_N$  the most / poorly constrained regions," but your answer focuses on decision-boundary/entropy ideas without tying to Bayesian LR informativeness.

**Q6(a) -0.5:** Plotting choices can be misleading (e.g., fixed data-space y-limits  $[-1, 1]$  may clip regression lines/data;  $N=0$  "likelihood" handling is not cleanly

justified).

**Q6(b) –1.0:** Multiple issues: (i) likelihood-shape description is inaccurate/over-specific and not tied to the constraint  $w_0 + w_1 x_1 \approx y_1$ ; (ii) “posterior center” is estimated via grid argmax (MAP on a grid) instead of reporting the posterior mean  $m_N$  as requested; (iii) uncertainty labels/descriptions are ad-hoc (prior should be clearly broad at  $N=0$ ).

**Q7(a) –0.25:** Good trends overall, but you incorrectly suggest precision growth looks the same “regardless of  $\sigma$ ”; in fact precision increases by  $1/\sigma^2$  per sample, so larger  $\sigma \Rightarrow$  slower precision growth.

**Q7(b) –0.25:** Correct direction (higher noise  $\Rightarrow$  more cautious updates), but the link from  $\sigma$  to the update mechanics is not stated cleanly (learning-rate/precision dependence).

**Q8(a) –0.5:** You simulate drifting true temperature and generate new readings, but the Bayesian update uses the *old* measurement stream  $x_t$  rather than the newly generated drifting measurements, so it doesn’t match the intended experiment.

**Q8(b) –0.5:** Main conclusion is broadly right (static-parameter learner becomes overconfident and can’t track drift), but reasoning contains irrelevant claims and does not clearly state the needed fix (a dynamic/state-space model or forgetting/transition model).

**Q9 –0.5:** You end with the correct-looking form, but the derivation maximizes a single-sample likelihood rather than the joint likelihood for  $N$  i.i.d. samples, and includes incorrect/irrelevant claims (e.g., concavity/bell-curve intuition).

**Q11(a) –1:** Posterior is not derived correctly for  $N$  samples; the correct Gamma–Exponential conjugate update is not obtained.

**Q11(b) –1:** You conclude the prior is not conjugate, which is incorrect; the given prior is Gamma and is conjugate to the exponential likelihood.

-Q12 : 2/4

- **Q12(a) -0.5:** You did **not** simulate lifetimes from **Exponential(rate =  $\theta_0 = 0.2$ )** for  $N=\{10,100\}$ . Instead you generate **Uniform** samples and even set  $\theta = -0.2$  as an “initial weight”, which is not the required exponential data simulation.
- **Q12(b) -0.5:** You wrote the *exponential* MLE formula  $\hat{\theta} = 1/\bar{x}$ , but you compute it on **Uniform(0,1)** samples (and the printed “true theta” is inconsistent), so the reported MLE is **not valid for the required exponential data**.
- **Q12(c) -0.5:** You define the prior  $p(\theta) = \theta^3 e^{-3\theta}$  correctly, but you build the **likelihood for a Uniform  $U(0,\theta)$**  model and then set  $\text{posterior} = \text{likelihood} * \text{prior}$  without deriving the required **Gamma posterior parameter update** for the exponential model.
- **Q12(d) -0.5:** Your conjugacy discussion is incorrect (you argue conjugacy fails based on “peaks/touching spikes” reasoning). For the intended model, the prior is **Gamma** and is **conjugate** to the exponential likelihood; your conclusion is not consistent with the theoretical conjugacy behavior.

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|------------|-----------|-----------|----|
| Assignment | 19.5 / 33 | 5.91 / 10 | D+ |
|------------|-----------|-----------|----|

2

Overall Feedback

Total score: 19.5/33

problem 1 total: 8.5/15 (with 1 points bonus)

problem 2 total: 11/18

### Deduction details:

**1.1d: -0.5:** Correlation analysis is incomplete. The submission reports small correlations with limited explanation, but does not clearly explain why  $w_1$  and  $w_2$  can be correlated.

**1.1f: -1.0:** The required coefficient trajectory plot during training is missing. Only final posterior means were shown, without convergence/stability analysis.

**1.2a: -1.0:** The Student-t model did not fix  $v=4$  as required. Instead,  $v$  was treated as a random variable.

**1.2d: -1.5:** The outlier experiment was implemented incorrectly. Instead of adding +12 to three randomly selected existing  $y_i$ , the submission created separate synthetic outlier points.

**1.2e: -0.5:** Predictive coverage was computed from intervals on  $\mu$  rather than proper posterior predictive intervals for  $y$ .

**1.3b: -0.5:** Posterior summaries are incomplete. Only posterior means were reported, while fuller summaries were expected.

**1.3c: -1.5:** This part is mostly missing. The required plot of  $s(x)$  with posterior median and 90% credible band was not provided.

**1.3d: -0.5:** The comparison with the homoscedastic baseline is too limited and does not clearly explain where uncertainty increases and why.

**1.3e: -0.5:** The discussion of identifiability/optimization challenges is only partial. It does not clearly explain that mean-model errors may be absorbed by the noise model.

**1.4a–1.4e: +1** The bonus part is only partially correct. Main issues include fixed noise instead of learning  $\sigma_\epsilon$ , incomplete prior setup, missing proper loss curve, incomplete posterior summaries, and an incorrect comparison baseline.

**2.1b: -1.0:** The answer does not clearly identify the correct cached forward-pass quantities needed for backpropagation.

**2.2b: -1.0:** The variance derivation is incorrect. The correct dependence on batch size is not properly derived.

**2.2c: -0.5:** The conclusion about doubling batch size is incorrect. Gradient noise should decrease, not stay unchanged.

**2.3b: -0.5:** The suggested interventions are only partially convincing and not as targeted as expected.

**2.4: -0.5:** One example is less precise and shifts away from leakage/evaluation issues toward a more general training bug.

**2.5a: -0.5:** The parameter-count derivation is incomplete, especially regarding bias terms and clear FC formula presentation.

**2.6b: -0.5:** The explanation of pooling invariance is vague and does not clearly distinguish stability from exact invariance.

**2.6c: -0.5:** The counterexample showing max-pooling is not exactly translation-equivariant is not clearly presented.

**2.7: -0.5:** The explanation mixes residual connections with U-Net-style skip connections, which makes the argument less precise.

**2.9a: -0.5:** The general forward-pass expression for the RNN is not written correctly.

**2.9b: -0.5:** The BPTT recurrence for  $\partial L / \partial h_t$  is incorrect.

**2.9c: -0.5:** The explanation of vanishing/exploding gradients is mathematically inaccurate in several places.

|           |          |      |
|-----------|----------|------|
| Midterm   | 70 / 100 | 10.5 |
| Exam      |          |      |
| 1 (Bonus) |          |      |

|         |         |          |   |
|---------|---------|----------|---|
| Midterm | 11 / 22 | 7.5 / 15 | D |
| Exam 2  |         |          |   |

|            |           |           |   |
|------------|-----------|-----------|---|
| Assignment | 28.5 / 32 | 8.91 / 10 | A |
| 3          |           |           |   |

Overall Feedback

Total score: 28.5/37

### Question1 -4 points

Problem1:

(e) -0.5: The explanation of Hungarian vs. row-wise softmax starts correctly, but the final statement about when one-to-one matching is essential is not accurate enough. In multi-target tracking, the one-to-one constraint is broadly important, not only in the narrower way described.

Problem2:

(e) -0.5: The answer correctly recognizes the failure pattern, but it does not clearly state the key desired gate behavior in the standard form, namely that the system should drive the radar weight toward zero and rely mainly on the camera embedding.

(f) -0.5: The answer gives the right high-level idea that three modalities need normalized weights summing to one, but the mathematical extension is not written cleanly. It mixes sigmoid-style notation with

the softmax requirement rather than specifying the softmax gate in a precise way.

Problem 3:

(a) -0.5: The answer recognizes that JPDA involves marginalization over joint events, but the explicit computation is not correct. The number of valid joint events is not worked out properly, and the marginal formulas are not normalized correctly.

(e) -0.5: This part contains several clear problems. The complexity discussion is inaccurate, the uncertainty comparison is muddled, and the final recommendation is inconsistent across the explanation.

Problem 4:

(a) -0.5: The design includes several relevant ideas, but it is not coherent enough as a single C-UAS solution. The association choice, temporal model discussion, and uncertainty section mix multiple incompatible directions, and some justifications are not well matched to the stated real-time design constraints.

(b) -0.5: The metric discussion contains some useful content, but the deployment argument is not clearly resolved around the assignment's intended latency trade-off. The answer does not cleanly settle on the correct practical deployment choice in the way expected.

## Question 2 -5 points

Problem 2:



(b) - 0.5: The answer identifies the denoiser inputs, but the explanation of why conditioning only once is weaker is not well aligned with the intended diffusion-based reasoning. It becomes overly speculative and drifts away from the core idea of needing conditional information at every reverse step.

(d) -1.0: This answer is clearly off. TimeGrad should be placed primarily in sequence modeling, but the response introduces set/graph-like dependence in a way that does not fit the intended taxonomy answer.

Problem3:

(a) -0.5: The answer knows uncertainty is not represented as a simple explicit parameter, but it becomes overly tied to internal noise-process notation instead of clearly answering that uncertainty is represented implicitly through sampling from the learned conditional distribution.

(b) -0.5: The requested outputs were a point estimate, a 90% interval, and a calibration diagnostic. Instead of answering them directly and cleanly, the response becomes tangled in MLE/MAP/credible-interval discussion and does not give the expected simple sample-based procedures.

(c) -1.0: This part is substantially incorrect. Aleatoric, epistemic, and Monte Carlo variability are not separated properly, and the answer incorrectly treats sampling variability as if it were directly captured by the model in the intended sense.

Problem 4

- (a) - 0.5: The answer points out that RMSE/MAE are point-estimate metrics, but much of the explanation is tied to unrelated gradient-descent reasoning rather than the core issue of missing distributional evaluation.
- (b) -0.5: The CRPS formula is written, and the answer gestures toward MAE and Brier, but the explanation is muddled and does not cleanly explain what CRPS rewards and penalizes in the intended probabilistic sense.
- (c) -0.5: Some relevant points are present, but the answers for CRPS suitability, CRPSSum, and the extra engineering evaluations are not clearly separated and not fully aligned with the question. The final engineering suggestions are also not especially well justified.